

**SAFETY DATA SHEET**

Creation Date 29-March-2019

Revision Date 06-May-2025

Revision Number 4

**1. Identification**

**Product Name** Bifenox

**Cat No. :** AC464600000, AC464602500, AC464600010

**CAS-No** 42576-02-3  
**Synonyms** Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Acute oral toxicity**

Category 4

**Label Elements****Signal Word**

Warning

**Hazard Statements**

Harmful if swallowed

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Response**

IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                                      | CAS-No     | Weight % |
|------------------------------------------------|------------|----------|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | <=100    |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Dry chemical, CO <sub>2</sub> , water spray or alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                                                                                                                       |
| <b>Flash Point</b>                    | No information available                                                                                                                                                       |
| <b>Method -</b>                       | No information available                                                                                                                                                       |
| <b>Autoignition Temperature</b>       | No information available                                                                                                                                                       |
| <b>Explosion Limits</b>               | No information available                                                                                                                                                       |

|                                  |                          |
|----------------------------------|--------------------------|
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen chloride.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

Health  
2

Flammability  
1

Instability  
0

Physical hazards  
N/A

**6. Accidental release measures**

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                                                           |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                                                                                                |

**7. Handling and storage**

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Store in freezer. Incompatible Materials. Strong oxidizing agents.                                                                                                                  |

**8. Exposure controls / personal protection**

|                                                                              |                                                                                                                                                                                                                                                                                                                                                                 |                             |                                                 |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------|
| <b>Exposure Guidelines</b>                                                   | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.                                                                                                                                                                                                                    |                             |                                                 |
| <b>Engineering Measures</b>                                                  | Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |                             |                                                 |
| <b>Personal protective equipment</b>                                         |                                                                                                                                                                                                                                                                                                                                                                 |                             |                                                 |
| <b>Eye Protection</b>                                                        | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                                                                                                                                     |                             |                                                 |
| <b>Hand Protection</b>                                                       | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                                                                                                                                       |                             |                                                 |
| <b>Glove material</b><br>Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | <b>Breakthrough time</b><br>See manufacturers recommendations                                                                                                                                                                                                                                                                                                   | <b>Glove thickness</b><br>- | <b>Glove comments</b><br>Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                               |
|----------------------------------------|-------------------------------|
| Physical State                         | Solid                         |
| Appearance                             | Yellow                        |
| Odor                                   | No information available      |
| Odor Threshold                         | No information available      |
| pH                                     | No information available      |
| Melting Point/Range                    | 84 - 89 °C / 183.2 - 192.2 °F |
| Boiling Point/Range                    | No information available      |
| Flash Point                            | No information available      |
| Evaporation Rate                       | Not applicable                |
| Flammability (solid,gas)               | No information available      |
| Flammability or explosive limits       |                               |
| Upper                                  | No data available             |
| Lower                                  | No data available             |
| Vapor Pressure                         | No information available      |
| Vapor Density                          | Not applicable                |
| Specific Gravity                       | No information available      |
| Solubility                             | Insoluble in water            |
| Partition coefficient; n-octanol/water | No data available             |
| Autoignition Temperature               | No information available      |
| Decomposition Temperature              | No information available      |
| Viscosity                              | Not applicable                |
| Molecular Formula                      | C14 H9 Cl2 N O5               |
| Molecular Weight                       | 342.13                        |

## 10. Stability and reactivity

|                                  |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                     |
| Stability                        | Stable under recommended storage conditions.                                                                   |
| Conditions to Avoid              | Incompatible products.                                                                                         |
| Incompatible Materials           | Strong oxidizing agents                                                                                        |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Hydrogen chloride |

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                                         | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation              |
|---------------------------------------------------|---------------------------|-------------------------------|------------------------------|
| Methyl<br>5-(2,4-dichlorophenoxy)-2-nitrobenzoate | LD50 = 6400 mg/kg ( Rat ) | LD50 > 20000 mg/kg ( Rabbit ) | LC50 > 0.91 mg/L ( rat ) 4 h |

**Toxicologically Synergistic** No information available

#### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                         | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Methyl<br>5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                                         | Freshwater Algae | Freshwater Fish                               | Microtox   | Water Flea           |
|---------------------------------------------------|------------------|-----------------------------------------------|------------|----------------------|
| Methyl<br>5-(2,4-dichlorophenoxy)-2-nitrobenzoate | Not listed       | LC50: >0.18 mg/l/96H<br>(Oncorhynchus mykiss) | Not listed | EC50: >0.35 mg/L/48H |

**Persistence and Degradability** May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component                                      | log Pow |
|------------------------------------------------|---------|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 4.48    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3077  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
 Technical Name Bifenox  
 Hazard Class 9  
 Packing Group III

#### TDG

UN-No UN3077  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
 Hazard Class 9  
 Packing Group III

#### IATA

UN-No UN3077  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
 Hazard Class 9  
 Packing Group III

#### IMDG/IMO

UN-No UN3077  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.  
 Hazard Class 9  
 Packing Group III

### 15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed Japan (ISHL) Japan (ISHL)

#### International Inventories

| Component                                      | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | -   | -    | -    | -                                             | 255-894-7 | -      | -   |

| Component                                      | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | -     | KE-10174 | -    | X    | X    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

Authorisation/Restrictions according to EU REACH Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                      | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                      | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Methyl 5-(2,4-dichlorophenoxy)-2-nitrobenzoate | 42576-02-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

Prepared By Regulatory Affairs  
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**Disclaimer**

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**End of SDS**